Chapter 15, Problem 2P

Problem

Determine the Laplace transform of 3.5 cos (5*t* − 45°).

Step-by-step solution

Step 1 of 6

Consider the following signal:

$$f(t) = 3.5\cos(5t - 45°)$$

It is known that,

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$$

Therefore,

$$\begin{aligned} f(t) &= 3.5[\cos 5t\cos 45° + \sin 5t\sin 45°] \\ &= 3.5\left[\cos 5t\left(\frac{1}{\sqrt{2}}\right) + \sin 5t\left(\frac{1}{\sqrt{2}}\right)\right] \\ &= \frac{3.5}{\sqrt{2}}(\cos 5t + \sin 5t) \end{aligned}$$

Comments (2)

Anonymous

Would be faster just to use a table at this point.

Anonymous

it helps if you're trying to learn the concept, but yeah the table is better

Step 2 of 6

The Laplace transform of signal $f(t)$ is,

$$L[f(t)] = \int_0^\infty f(t)e^{-st}\,dt$$

$$F(s) = \int_0^\infty f(t)e^{-st}\,dt$$

Substitute $\frac{3.5}{\sqrt{2}}(\cos 5t + \sin 5t)$ for $f(t)$ in the equation.

$$\begin{aligned} F(s) &= \int_0^\infty \frac{3.5}{\sqrt{2}}(\cos 5t + \sin 5t)e^{-st}\,dt \\ &= \frac{3.5}{\sqrt{2}}\int_0^\infty (\cos 5t + \sin 5t)e^{-st}\,dt \end{aligned}$$

$$F(s) = \frac{3.5}{\sqrt{2}}\int_0^\infty \cos(5t)e^{-st}\,dt + \frac{3.5}{\sqrt{2}}\int_0^\infty \sin(5t)e^{-st}\,dt \quad \ldots\ldots (1)$$

Step 3 of 6

Simplify the expression $\int_0^\infty \cos(5t)e^{-st}\,dt$.

It is known that,

$$\cos(\omega t)=\frac{e^{j\omega t}+e^{-j\omega t}}{2}$$

Therefore,

$$\begin{aligned}\int_0^\infty \cos(5t)e^{-st}\,dt &= \int_0^\infty\left(\frac{e^{j5t}+e^{-j5t}}{2}\right)e^{-st}\,dt\\ &=\frac{1}{2}\left[\int_0^\infty e^{j5t}\,e^{-st}\,dt+\int_0^\infty e^{-j5t}\,e^{-st}\,dt\right]\\ &=\frac{1}{2}\left[\int_0^\infty e^{-(s-j5)t}\,dt+\int_0^\infty e^{-(s+j5)t}\,dt\right]\\ &=\frac{1}{2}\left[\left[\frac{e^{-(s-j5)t}}{-(s-j5)}\right]_0^\infty+\left[\frac{e^{-(s+j5)t}}{-(s+j5)}\right]_0^\infty\right]\end{aligned}$$

Further simplification as follows,

$$\begin{aligned}\int_0^\infty \cos(5t)e^{-st}\,dt &=\frac{1}{2}\left[\left[\frac{e^{-(s-j5)(\infty)}}{-(s-j5)}-\frac{e^{-(s-j5)(0)}}{-(s-j5)}\right]+\left[\frac{e^{-(s+j5)(\infty)}}{-(s+j5)}-\frac{e^{-(s+j5)(0)}}{-(s+j5)}\right]\right]\\ &=\frac{1}{2}\left[\left[\frac{0}{-(s-j5)}-\frac{1}{-(s-j5)}\right]+\left[\frac{0}{-(s+j5)}-\frac{1}{-(s+j5)}\right]\right]\\ &=\frac{1}{2}\left[\frac{1}{(s-j5)}+\frac{1}{(s+j5)}\right]\end{aligned}$$

$$=\frac{1}{2}\left[\frac{s+j5+s-j5}{(s-j5)(s+j5)}\right]$$

$$=\frac{1}{2}\left[\frac{2s}{s^2+25}\right]$$

$$=\frac{s}{s^2+25}$$

Step 4 of 6

Simplify the expression $\int_0^\infty \sin(5t)e^{-st}\,dt$.

It is known that,

$$\sin(\omega t)=\frac{e^{j\omega t}-e^{-j\omega t}}{2j}$$

Therefore,

$$\begin{aligned}\int_0^\infty \sin(5t)e^{-st}\,dt &= \int_0^\infty\left(\frac{e^{j5t}-e^{-j5t}}{2j}\right)e^{-st}\,dt\\ &=\frac{1}{2j}\left[\int_0^\infty e^{j5t}\,e^{-st}\,dt-\int_0^\infty e^{-j5t}\,e^{-st}\,dt\right]\\ &=\frac{1}{2j}\left[\int_0^\infty e^{-(s-j5)t}\,dt-\int_0^\infty e^{-(s+j5)t}\,dt\right]\\ &=\frac{1}{2j}\left[\left[\frac{e^{-(s-j5)t}}{-(s-j5)}\right]_0^\infty-\left[\frac{e^{-(s+j5)t}}{-(s+j5)}\right]_0^\infty\right]\end{aligned}$$

Step 5 of 6

Further simplification as follows,

$$\int_0^{\infty} \sin(5t)e^{-st}\,dt = \frac{1}{2j}\left[\left[\frac{e^{-(s-j5)(\infty)}}{-(s-j5)} - \frac{e^{-(s-j5)(0)}}{-(s-j5)}\right] - \left[\frac{e^{-(s+j5)(\infty)}}{-(s+j5)} - \frac{e^{-(s+j5)(0)}}{-(s+j5)}\right]\right]$$

$$= \frac{1}{2j}\left[\left[\frac{0}{-(s-j5)} - \frac{1}{-(s-j5)}\right] - \left[\frac{0}{-(s+j5)} - \frac{1}{-(s+j5)}\right]\right]$$

$$= \frac{1}{2j}\left[\frac{1}{(s-j5)} - \frac{1}{(s+j5)}\right]$$

$$= \frac{1}{2j}\left[\frac{s+j5-s+j5}{(s-j5)(s+j5)}\right]$$

$$= \frac{1}{2j}\left[\frac{10j}{s^2+25}\right]$$

$$= \frac{5}{s^2+25}$$

Step 6 of 6

Recall equation (1),

$$F(s) = \frac{3.5}{\sqrt{2}}\int_0^{\infty}\cos(5t)e^{-st}\,dt + \frac{3.5}{\sqrt{2}}\int_0^{\infty}\sin(5t)e^{-st}\,dt$$

Substitute $\frac{s}{s^2+25}$ for $\int_0^{\infty}\cos(5t)e^{-st}\,dt$ and $\frac{5}{s^2+25}$ for $\int_0^{\infty}\sin(5t)e^{-st}\,dt$ in the equation.

$$F(s) = \frac{3.5}{\sqrt{2}}\left(\frac{s}{s^2+25}\right) + \frac{3.5}{\sqrt{2}}\left(\frac{5}{s^2+25}\right)$$

$$= \frac{3.5}{\sqrt{2}}\left[\frac{s+5}{s^2+25}\right]$$

Thus, the Laplace transform of $3.5\cos(5t-45°)$ is, $\boxed{\frac{3.5}{\sqrt{2}}\left[\frac{s+5}{s^2+25}\right]}$.